

MOST HAPPENING, FACT-FILLED CURRENT AFFAIRS FROM AROUND THE WORLD

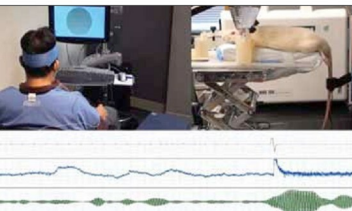
Supercomputers cannot match human brain

According to a Japan-based research, it has been discovered that even the fastest computers in the world cannot match the human brain. The Japanese researchers used the K Supercomputer, which is one of the fastest supercomputers in the world, to replicate the complicated neural structure of the human brain.

As stated on crazyengineers.com, the researchers used a neuron simulating software (NEST) and the power of 82,944 processors to create a network which is similar to 1.73 billion nerve cells joined by 10.4 trillion synapses. The research was carried out by the HPCI Programme for Computational Life Sciences at the Okinawa Institute of Technology Graduate University in Japan. It was eventually found that even with the help of a top supercomputer, the researchers could attain only one per cent of the raw processing power of the brain.

Brain-to-brain interface from Harvard

The first non-invasive brain-to-brain interface (BBI) has been created by researchers at Harvard University. The interface is designed to allow a human to control an animal through his thoughts.



Harvard researchers' brain-to-brain interface allows humans to control other animals through thoughts



The University of Tokyo robot engineer, Tomotaka Takashi chats with Kirobo (right) and its backup robot, Mirata, in Tokyo

The BBI interface functions by making the human wear a EEG-based BCI. A mouse on the other hand is made to wear a focused ultrasound (FUS) computer-brain interface (CBI). The FUS allows the researchers to excite a particular region of neurons in the rat's brain through an ultrasound signal.

Once things are in place, the BCI reacts as and when the human looks at a specific pattern on a computer screen. As the BCI sends a command to the rat's CBI, an ultrasound is beamed into the rat's motor cortex that controls its tail movement. Researchers feel that the human BCI is as accurate as 94 per cent, and takes 1.5 seconds for the process to complete. The experiment's success will make way for sending across more complex ideas onto more complex beings.

Robot reaches space station

Japan has recently launched a cargo-carrying rocket, which was loaded with supplies meant for the crew of the International Space Station (ISS). The most interesting part was that the cargo also carried a small robot.

The small talking robot, Kirobo, which was sent for the mission, is the first Android object to reach the space and is designed to act as a talking companion for Koichi Wakata, who is one of the astronauts working on the ISS. Just 34cm tall and 1 kg in weight, Kirobo is programmed to converse in Japanese and record the conversations with Wakata, who is also the first-ever Japanese astronaut to work on the ISS.

Skilled robots soon

As per an international expert, soon robots will compete with human beings in capabilities and flexibility in performing difficult tasks.

The robotics and autonomous systems (RAS) technology has already had